EVERY ELEMENT OF A PROJECTIVELY INFINITE-RANK MODULE OVER A RING WITH NOETHERIAN RADICAL QUOTIENT LIES IN A FINITE STABLY FREE DIRECT SUMMAND

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 39107-39114 (canonical tag 0GVJ) and the counted proof is the proof environment at source lines 39116-39150, which immediately follows it and is its own proof.

Editorial note. The proof contains one footnote at source lines 39134-39141 verifying that the projection onto K" is surjective. It is the author's own text inside the counted span and is retained verbatim.

Editorial note. The two lemmas the proof invokes, lemma-element-projective and lemma-trick-to-find-good-element, are earlier statements of the same section of this chapter and are carried as navigation stubs with their roles recorded in the reasons.

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Allowed standard prerequisites (editorial).

- An element of a projective module that is not a zero divisor on the fibres generates a free direct summand after one passes to a suitable element of the module. invoked at proof lines 39126-39129, where the surjection onto the complement K' is used to find p with the image of (1, p) generating a free direct summand. Produces the free summand that is split off from the complement inside R direct sum P.
- A projective module that is the kernel of a surjection onto a projective module is a direct summand, and being a direct summand of a projective module, hence projective. used at proof lines 39142-39144. Makes the kernel of the projection onto K" a direct summand of P containing the given element s.
- Stable freeness is preserved by taking direct summands of finite free modules: if R direct sum N is isomorphic to a finite free module then N is stably

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free. used at proof lines 39144-39149. Concludes that the direct summand produced is finite stably free, which is the conclusion of the lemma.

Source reference. Original lemma 0GVH: sources/stacks-more-algebra/pinned-original-source.tex, source lines 39009–39009. The reference supplies the auxiliary statement that places (0, s) inside a finite free direct summand of a finite free direct summand of R direct sum P, used at proof lines 39117-39120; the reduction and everything after it are inside the counted span.

Source reference. Original lemma 0GVI: sources/stacks-more-algebra/pinned-original-source.tex, source lines 39026–39026. The reference supplies the element p whose image generates a free direct summand of the complement, used at proof lines 39127-39130; the decomposition of the complement and the stable freeness conclusion are inside the counted span.

Lemma 0.2. *Let R be a ring with Jacobson radical J such that R/J is Noetherian. Let P be a projective R-module such that $P_{\mathfrak{m}}$ has infinite rank for all maximal ideals $\mathfrak{m}$ of R. Let $s \in P$. Then we can find a finite stably free direct summand $M \subset P$ such that $s \in M$.*

Proof. By Lemma 0GVH we can find a finite free module F and a finite free direct summand $K \subset F \oplus P$ such that $(0, s) \in K$. By induction on the rank of F we reduce to the case discussed in the next paragraph.

Assume there exists a finite stably free direct summand $K \subset R \oplus P$ such that $(0, s) \in K$. Choose a complement K' of K, i.e., such that $R \oplus P = K \oplus K'$. The projection $\pi : R \oplus P \to K'$ is surjectve, hence by Lemma 0GVI we find a $p \in P$ such that $\pi(1, p) \in K'$ generates a free direct summand. Accordingly we write $K' = R\pi(1, p) \oplus K''$. We see that

$$R \oplus P = K \oplus K' = K \oplus R\pi(1, p) \oplus K''$$

The projection $\pi' : P \to K''$ is surjective[1] and hence split (as K'' is projective). Thus $\mathrm{Ker}(\pi') \subset P$ is a direct summand containing s. Finally, by construction we have an isomorphism

$$R \oplus \mathrm{Ker}(\pi') \cong K \oplus R\pi(1, p)$$

and hence since K is finite and stably free, so is $\mathrm{Ker}(\pi')$. □

[1]Namely, if $k'' \in K''$ then k'' viewed as an element of K' can be written as $k'' = \lambda\pi(1, 0) + \pi(0, q)$ for some $\lambda \in R$ and $q \in P$. This means $k'' = \lambda\pi(1, p) + \pi(0, q - \lambda p)$. This in turn means that $q - \lambda p$ maps to k'' by the composition $P \to R \oplus P \xrightarrow{\pi} K' \to K''$ since $K' \to K''$ annihilates $\pi(1, p)$.